

Functional Requirements Document (FRD)

Inventory Management System — Bag & Shoes Retail Shop

Version: 1.0 **Status:** Initial Development Specification **Platform:** Web Application

Scope: Retail inventory and sales management. Barcode/scanner functionality is intentionally excluded from Version 1.0, but the database should remain barcode-ready for future implementation.

1. User Roles

Role	Access / Responsibilities
Owner	Full system access: dashboard, products, inventory, purchases, sales/POS, customers, suppliers, expenses, reports, u
Staff	Operational access: POS/sales, customers, product lookup, inventory operations, purchase receiving and stock counts,

2. Dashboard

- Display today's sales, transaction count, gross profit, monthly sales, monthly profit, total inventory quantity/value, low-stock products, out-of-stock products, supplier payables, customer receivables and recent transactions.
- Provide daily/weekly/monthly sales summaries and sales by payment method and category.
- Show inventory alerts for low-stock and out-of-stock products.

3. Product Management

- Create, edit, view and deactivate products.
- Product fields: product ID, name, SKU/product code, category, brand, description, image, supplier, purchase price, selling price, discount price, minimum stock level, status, created/updated date.
- Categories and brands must be configurable.

4. Product Variants

- Support shoe sizes and colors and bag colors/designs/materials.
- Each variant should have its own SKU, purchase price, selling price, quantity, minimum stock level and status.
- Example: Nike Air Max — Black — Size 42 — Stock 8.

5. Inventory Management

- Track opening stock, current stock, sold quantity, purchased quantity, returned quantity, damaged stock and other adjustments.
- Allow authorized users to create stock adjustments with reason.
- Record every inventory change in a stock movement history.
- Movement types: opening stock, purchase, sale, customer return, supplier return, damage, loss and manual adjustment.

6. Stock Count

- Allow staff/owner to perform physical stock counts.
- Show system quantity, physical quantity and difference.
- Require a reason before applying an adjustment and preserve the count history.

7. Purchase Management

- Create purchases with supplier, date, products, quantities, purchase price, discount, other costs, total and payment status.

- Purchase status: Draft, Ordered, Partially Received, Received, Cancelled.
- Receiving goods increases inventory only after quantities are confirmed.

8. Supplier Management

- Store supplier name/company, phone, email, address, PAN/VAT number, bank details, payment terms, notes and status.
- Track purchases, paid amount, outstanding amount and supplier payment history.

9. Sales / POS

- Allow product selection by name, SKU, category, brand, size and color.
- Cart must support quantity, unit price, discount and total.
- Calculate subtotal, discount, tax/charges if configured and grand total.
- Payment methods: Cash, eSewa, Khalti, Fonepay, Bank Transfer, Card, Credit and configurable Other.

10. Invoice Management

- Generate invoice after completed sales.
- Invoice includes shop details, invoice number, date/time, staff member, customer, items, quantities, prices, discounts, total, payment method, amount paid and balance.
- Invoice should be printable and downloadable as PDF.

11. Sales Returns & Exchanges

- Process returns against an original sale.
- Record return quantity, reason and refund/store-credit method.
- Returned stock must be added back to inventory where applicable.
- Support size/product exchanges and automatically calculate additional payment or refund.

12. Customer Management

- Store customer name, phone, email, address and notes.
- Display purchase history, total spending, number of orders, returns and outstanding balance.
- Support guest checkout.

13. Customer Credit

- Track credit invoices, amount paid, outstanding balance, payment history and due dates.
- Show customer receivables on dashboard and reports.

14. Expense Management

- Record rent, electricity, internet, salary, transportation, marketing, maintenance, packaging and miscellaneous expenses.
- Store category, amount, date, payment method, description and optional receipt/attachment.

15. Profit Calculation

- Calculate revenue, cost of goods sold, gross profit, expenses and net operating profit.
- Profit calculations must use the actual purchase cost associated with sold products/variants.

16. Reports

- Sales: daily, weekly, monthly, custom date range, product, category, brand, staff and payment method.
- Inventory: current stock, low stock, out of stock, stock valuation, stock movement, damaged items, adjustments, fast-moving and slow-moving products.
- Purchases: supplier, date and purchase amount.
- Financial: revenue, gross profit, expenses, net profit, customer receivables and supplier payables.
- Reports should support PDF and Excel export.

17. Low-Stock Management

- Allow a minimum stock level per product variant.
- Generate low-stock alerts when current stock reaches the configured threshold and out-of-stock alerts when quantity reaches zero.

18. Search & Filtering

- Products: name, SKU, brand, category, size and color.
- Sales: invoice number, customer, staff, date and payment method.
- Purchases: supplier, purchase number, date and status.

19. User & Permission Management

- Owner can create, deactivate and manage Staff accounts.
- Owner controls Staff permissions.
- Record user name, username/email, phone, role, status and last login.
- Staff access should be limited to operational functions required for daily shop work.

20. Audit Log

- Record critical actions such as stock adjustments, price changes, sales cancellation, returns, purchase changes, expense changes and user/permission changes.
- Each log should contain user, action, affected record, previous value where relevant, new value where relevant, reason where applicable and timestamp.

21. System Settings

- Shop name, logo, address, phone, email and PAN/VAT information.
- Invoice numbering and footer.
- Default discounts, taxes, payment methods and return policy.
- Inventory thresholds and staff permissions.

22. Non-Functional Requirements

- Use HTTPS and secure password hashing.
- Implement role-based access control.
- Provide automatic database backups.
- Support desktop, laptop, tablet and mobile responsive layouts.
- Normal POS and dashboard operations should target approximately 1–2 second response time under expected shop workload.

23. Version 1 Scope Exclusions

- Barcode scanner integration
- Barcode generation/printing
- Barcode label management
- AI demand forecasting
- Multi-branch management
- Online-store integration
- SMS/WhatsApp automation
- Advanced loyalty program

24. Future Barcode Readiness

- Although barcode scanning is excluded from Version 1, ProductVariant should include an optional barcode field so barcode functionality can be added later without redesigning the core inventory structure.

Primary business flow: Supplier → Purchase → Receive Goods → Inventory → Sale/POS → Payment → Invoice → Profit/Reports. Returns, exchanges and stock adjustments update inventory through controlled stock movements.